

# DengarTrack

Digital Newborn Hearing Screening Management Platform

## 1. Executive Summary

DengarTrack is a purpose-built mobile digital workflow platform designed to replace the fragmented paper-based management system currently used in Universal Newborn Hearing Screening (UNHS) programmes across Malaysian public hospitals. No equivalent MoH-sanctioned platform currently exists.

This proposal is addressed to an ICT professional or software development team being considered as a technical partner for the development of DengarTrack. It describes the product vision, technical requirements, scope of work, deliverables, timeline, and proposed engagement model.

### The Problem in Numbers

29% mean loss-to-follow-up (LTFU) rate across 48 Malaysian public hospitals -- nearly 3x the international benchmark of <=10%

Inter-hospital case transfers currently conducted via consumer WhatsApp groups -- a direct violation of the Personal Data Protection Act 2010 (PDPA)

Monthly MoH reporting requires 2-4 hours of manual data aggregation per coordinator, per hospital, every month

*Evidence base: published in Journal of Medical Screening (SAGE, March 2026) + qualitative fieldwork across 19 hospitals*

## 2. Product Overview

### 2.1 What DengarTrack Does

DengarTrack replaces the current UNHS workflow -- paper cards, Excel spreadsheets, WhatsApp groups, and manual MoH reports -- with a single integrated mobile platform. Every screener action at the cot side flows automatically into the coordinator dashboard, follow-up queue, and national reporting system.

Concept:

Design principles: Speed First (any screener action <60 seconds) | Offline-Capable (works without WiFi) | Bahasa Melayu/English bilingual | PDPA 2010 compliant | Android 9+ | One entry, many uses (data entered once by screener auto-populates dashboard, follow-up queue, and MoH report)

### 2.2 Six Core Modules

M1

Months 1-6

- \* Digital OAE/AABR (test) result recording at point of care
- \* QR wristband barcode scanning (babies ID)

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <ul style="list-style-type: none"> <li>* offline-capable with auto-sync</li> <li>* ward tablet and mobile phone primary interface</li> <li>* bilingual BM/English</li> <li>* under 60 seconds per entry</li> </ul>                                                                                                                                                                                                                                                                  |
| <b>M2</b><br>Months 4-9   | <ul style="list-style-type: none"> <li>* Automated referral engine with real-time status tracking</li> <li>* follow-up queue for coordinators</li> <li>* automated WhatsApp/SMS reminders to caregivers (Day 7 and Day 14)</li> <li>* Amber flag at Day 14 if no response</li> <li>* Red flag at Day 28</li> <li>* LTFU flag at Day 60</li> <li>* all actions timestamped and auditable</li> </ul>                                                                                  |
| <b>M3</b><br>Months 6-12  | <ul style="list-style-type: none"> <li>* Hospital coordinator dashboard with real-time programme visibility</li> <li>* live coverage rate and refer/LTFU counts</li> <li>* 1-3-6 benchmark tracker (screen by 1 month diagnose by 3 intervene by 6)</li> <li>* 6-month trend chart</li> <li>* ward-by-ward breakdown</li> <li>* hospital admin summary view with donut chart and comparison to national benchmark</li> </ul>                                                        |
| <b>M4</b><br>Months 8-12  | <ul style="list-style-type: none"> <li>* Automated KKM-format MoH monthly report generator</li> <li>* auto-populated from existing screening data</li> <li>* data completeness check before export</li> <li>* one-click Excel (.xlsx) and PDF export</li> <li>* submission log with date and exporter name</li> <li>* replaces 2-4 hours of manual monthly work</li> </ul>                                                                                                          |
| <b>M5</b><br>Months 10-18 | <ul style="list-style-type: none"> <li>* Secure encrypted inter-hospital handover system</li> <li>* replaces illegal WhatsApp clinical data sharing</li> <li>* Baby System ID only -- no names or IC numbers transmitted</li> <li>* encrypted in-app communication with full audit trail</li> <li>* automated escalation if no response within 72 hours</li> <li>* PDPA 2010 fully compliant</li> <li>* supports 1-3-6 continuity when babies transfer between hospitals</li> </ul> |
| <b>M6</b><br>Months 12-18 | <ul style="list-style-type: none"> <li>* National KKM registry integration</li> <li>* hospital information system (HIS/iPerak) hooks</li> <li>* JPN MyKad integration for baby ID linkage</li> <li>* national-level aggregate dashboard for MoH data officers</li> <li>* FHIR R4 compliant data exchange</li> </ul>                                                                                                                                                                 |



## 3. Technical Requirements

### 3.1 Technology Stack

**IMPORTANT:** The technology stack below is the recommended architecture based on the PDPA compliance requirements, KKM hospital device inventory, and offline-first design constraint. The developer is invited to propose alternatives with justification.

| Layer                | Specification                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------|
| Frontend -- Mobile   | React Native (cross-platform Android/iOS) OR Flutter -- Android primary target, iOS secondary            |
| Frontend -- Web      | React.js -- for coordinator portal and KKM reporting dashboard on desktop/laptop                         |
| Backend              | Node.js with Express OR FastAPI (Python) -- RESTful API architecture                                     |
| Database             | PostgreSQL -- primary relational database for patient records, screening results, follow-up tracking     |
| Offline storage      | SQLite (mobile) + IndexedDB (web) -- offline-first with background sync when connectivity restored       |
| Authentication       | JWT tokens + Role-Based Access Control (RBAC) -- 4 roles: Screener, Coordinator, Hospital Admin, MoH/KKM |
| Cloud hosting        | AWS Malaysia region (ap-southeast-1) OR Azure Malaysia -- data residency requirement for PDPA compliance |
| Messaging            | WhatsApp Business API (Twilio) + SMS fallback (Twilio or Nexmo) -- for automated parent reminders        |
| Barcode scanning     | ZXing.js (web) OR React Native Camera (mobile) -- for QR wristband scanning at point of care             |
| Report export        | SheetJS -- Excel (.xlsx) export for KKM reports                                                          |
| Health data standard | HL7 FHIR R4 -- for KKM registry and HIS integration (Phase 2)                                            |
| Encryption           | TLS 1.3 in transit + AES-256 at rest -- mandatory for PDPA compliance                                    |

### 3.2 User Roles and Access Control

| Role                      | Access level                                                                                                    |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| Screener                  | Can enter and view own screening results only. Cannot access other wards, coordinator data, or reports.         |
| Coordinator (Audiologist) | Full access to their hospital -- dashboard, follow-up queue, MoH reports, handover initiation and receipt.      |
| Hospital Admin            | Hospital-level summary metrics and audit logs only. Cannot view individual baby records.                        |
| MoH / KKM                 | Aggregate national dashboard and national report download only. Cannot access individual hospital or baby data. |

### 3.3 Security and PDPA Compliance Requirements

The following are non-negotiable requirements, not optional features:

- \* All patient data encrypted in transit (TLS 1.3) and at rest (AES-256)
- \* Baby identifiers anonymised at inter-hospital transfer level -- Baby System ID only, never names or IC numbers transmitted externally
- \* Full immutable audit log -- every action timestamped with user ID. Append-only, cannot be edited or deleted.
- \* Role-based access enforced at API level, not just front-end
- \* No clinical data transmitted via any external consumer platform (WhatsApp, email, etc.)
- \* Data residency in Malaysia -- no data stored outside Malaysia region
- \* PDPA Data Protection Officer consultation required before pilot deployment
- \* External cybersecurity penetration test required before hospital pilot

### 3.4 Device and Connectivity Requirements

| Requirement        | Specification                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------|
| Primary device     | Android tablet (8-10 inch) and Android smartphone -- Android 9 and above (covers 94% of KKM ward devices) |
| iOS support        | Secondary -- required for full market coverage but not blocking for Pilot 1                               |
| Offline capability | Full core screening entry and shift summary must work with zero internet connectivity                     |
| Sync behaviour     | Auto-sync when WiFi or mobile data available. Show sync status indicator to screener.                     |
| Screen resolution  | Must be readable and usable on 720p Android devices                                                       |
| Language           | Bahasa Melayu as default. English toggle. All system messages and automated SMS in BM.                    |
| Accessibility      | WCAG AA contrast ratios -- critical for clinical environments with poor lighting                          |

## 4. Scope of Work

### 4.1 What the Developer Is Responsible For

- \* Full-stack mobile and web application development across all 6 modules
- \* Database design, setup, and migration scripts
- \* API development and documentation
- \* PDPA-compliant data architecture and security implementation
- \* AWS/Azure Malaysia region infrastructure setup and configuration
- \* WhatsApp Business API and SMS gateway integration
- \* QR barcode scanning integration
- \* Excel and PDF report export implementation
- \* Offline sync mechanism development
- \* Bug fixing and technical issue resolution throughout pilot studies

- \* Code documentation and handover package at project close
- \* Unit testing for all core modules

## 4.2 What the Researcher Is Responsible For

- \* All clinical domain expertise and product requirements
- \* Figma prototype and all screen-by-screen UI/UX specifications (already completed for 7 of 10 screens)
- \* Hospital access, coordinator relationships, and pilot study coordination
- \* NMRR ethics amendment for each pilot phase
- \* MoU facilitation between developer and UKM/hospitals
- \* User acceptance testing coordination with hospital staff
- \* Content for all bilingual Bahasa Melayu/English screen labels and SMS templates
- \* Final sign-off on clinical workflow logic and KKM report format
- \* All academic publications and conference presentations

## 4.3 What Is Out of Scope

- \* Hardware procurement (tablets, phones) -- hospital responsibility
  - \* Hospital WiFi infrastructure -- hospital responsibility
  - \* KKM registry API development -- KKM/MOH responsibility (Phase 2)
  - \* App Store / Play Store submission and ongoing maintenance after project close -- to be negotiated separately
  - \* Training of hospital staff -- researcher responsibility
-

## 5. Phased Development Plan

Development is structured across 4 phases aligned with the PhD pilot study timeline. Grant funding covers Phases 1-3.

| Phase                               | Timeline                         | Deliverables and Milestones                                                                                                                                                                                                                                                  |
|-------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1 Validate &amp; Setup</b> | Months 1-3 Apr - Jun 2026        | Development environment setup. Review Figma prototype with researcher. Finalise database schema and API contract. Hospital coordinator validation sessions (researcher-led, developer attends 1). Technical architecture document finalised. Agreed sprint plan for Phase 2. |
| <b>Phase 2 Build MVP</b>            | Months 4-9 Jul - Dec 2026        | M1 Point-of-Care Entry -- fully functional (offline + sync). M2 Follow-Up Engine -- manual queue + alert system. M3 Coordinator Dashboard -- live data from M1/M2. Basic admin portal. Internal QA testing. Ready for Pilot 1 (usability) by October 2026.                   |
| <b>Phase 3 Pilot Ready</b>          | Months 10-18 Jan - Sep 2027      | M2 automated WhatsApp/SMS reminders live. M4 MoH Report Generator. M5 Secure Inter-Hospital Handover. Penetration test + PDPA audit. Live in 2-3 pilot hospitals (Pilot 1: Oct-Dec 2026, Pilot 2: Jan-Jun 2027). Bug fixes from pilot feedback.                              |
| <b>Phase 4 Scale</b>                | Months 19-24 Oct 2027 - Mar 2028 | M6 KKM national registry integration. National rollout portal. HIS/iPerak hooks. Performance optimisation for 48-hospital scale. ASEAN market adaptation. Final code handover and documentation package.                                                                     |

### 5.1 Phase 1 Immediate Priorities (April - June 2026)

Given the i-ACHIEVE 2026 competition on 16 June 2026, the following must be completed in Phase 1:

1. Review the completed Figma prototype (7 of 10 screens built) and confirm technical feasibility of all specifications
2. Complete the remaining 3 Figma screens (5A Initiate Handover, 5B Handover Inbox, 5C Audit Record) in collaboration with researcher
3. Set up development environment and GitHub repository
4. Build a functional demonstration of M1 screening entry on Android -- live data entry that saves to a database and displays in a simple list
5. Prepare demonstration for i-ACHIEVE 2026 on 16 June 2026 at MRANTI Park Hall, Kuala Lumpur

## 6. Current Development Status

The following has already been completed by the researcher prior to engaging a developer:

| Item                       | Status                                                            |
|----------------------------|-------------------------------------------------------------------|
| <b>Problem validation</b>  | COMPLETE -- Published national evaluation (JMS, SAGE, March 2026) |
| <b>User needs analysis</b> | COMPLETE -- 19-hospital qualitative fieldwork, CFIR-guided        |

|                              |                                                                              |
|------------------------------|------------------------------------------------------------------------------|
| <b>Ethics approval</b>       | COMPLETE -- NMRR Amendment No. 1 approved                                    |
| <b>Technical brief</b>       | COMPLETE -- 10-section App Outline v1.1 with screen-by-screen specifications |
| <b>Figma Design System</b>   | COMPLETE -- Colour styles, text styles, 6 reusable components built          |
| <b>Figma Screens M1</b>      | COMPLETE -- Screens 1A Login, 1B Screening Entry, 1C Shift Summary           |
| <b>Figma Screens M2-M3</b>   | COMPLETE -- Screen 2A Follow-Up Queue, 3A Dashboard built; 3B and 4A partial |
| <b>Figma Screens M5</b>      | IN PROGRESS -- 5A, 5B, 5C to be completed                                    |
| <b>Prototype connections</b> | PENDING -- Screen flow connections to be linked                              |
| <b>Grant applications</b>    | IN PROGRESS -- Cradle CIP 500 (RM 500k) and MoH Innovation Grant             |
| <b>Competition entry</b>     | IN PROGRESS -- i-ACHIEVE 2026 abstract submitted, presentation 16 June 2026  |

*Note: Grant funding is not yet confirmed. Phase 1 development (April-June 2026) may need to proceed on a goodwill / equity basis pending grant outcome. This is standard for early-stage health tech ventures.*

## 6.1 IP and Publication Agreement

- \* All code developed for DengarTrack will be jointly owned by the researcher and developer (Option A) or fully owned by UKM (Option B)
- \* Developer will be acknowledged in all academic publications and conference presentations
- \* A formal Intellectual Property Agreement will be executed before any development work begins
- \* UKM's Research, Innovation, Commercialisation and Consultancy Management Centre (RICCC) will be engaged for IP registration

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